

TRANSERIES / CRITICAL INVERSION / COUNTABLE EXPONENTIAL FEEDBACK

Marginal Critical Transseries

Convergent Lambert– W Charts,
Universal Cutoff Corrections,
and Certified Action Budgets

Research draft prepared for Vladimir Reshetnikov

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Abstract

A countable exponential-action family with an exact j^{-3} tail reaches a critical boundary that is neither an analytic fold nor a nonintegral stable power singularity. We prove a convergent inverse chart around a Lambert– W_{-1} core, with rational coefficient functions, controlled support, and geometric remainders. We then obtain an all-orders Gaussian–logarithmic expansion of the original exponential-sector coefficients in a moving coupling window. The principal cutoff theorem separates two scales: fluctuations have size $\sqrt{n \log n}$, but retaining the n th sector to fixed relative accuracy requires only $\sqrt{n}$ actions. At a comparable cutoff the exact limiting retained fraction is $e^{-1/(2\ell^2)}$. Its first two-scale correction is independent of every finite change in the action weights, after normalization. At critical coupling this yields a two-term asymptotic formula for the minimal action budget. An explicit finite- n inequality, based on positive coefficient recurrences and an elementary Fourier majorant, supplies a separate genuine error certificate. Finally, a convergent reduced logarithmic chart for the finite-action fold explains how those folds approach the marginal boundary. Conventional proofs, exact finite checks, numerical diagnostics, and further research targets are provided.

Research status. The results concern a precisely specified positive, eventually exact power-tail class. The classical mechanisms—Lagrange inversion, Lambert inversion, normal attraction, and Poisson extremes—are not claimed as new. The proposed contribution is the combined explicit marginal theory, particularly the prefix-independent cutoff correction and its connection to a finite error certificate. Global publication priority has not been established. No new Lean formalization or independently reviewed proof is claimed.

Editorial note (ProveIt, 2026-09-29). “Certified” in the title refers to the proved inequality of Theorem 6.2. The recorded numerical evaluations use no directed rounding and are not interval or proof-assistant certificates; Section 10, Question 7, keeps such an implementation open.

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1 Research gap, contribution, and provenance

1.1 The question left by finite-action and stable-action models

The inspected **ProveIt** material contains an extensive inversion calculus, a finite-action moving-fold analysis, and a classification of countable exponential feedback [1, 2, 3]. These address different phenomena. Analytic finite-action sums can produce a square-root critical inverse. A countable sum may instead reach its own nonanalytic boundary before an analytic critical-point theorem applies. The companion draft *Beyond Finite-Action Folds* treats the exact tail $a_j \asymp j^{-1-\alpha}$ with $1 < \alpha < 2$, including a stable-scale cutoff law [4]. The endpoint $\alpha = 2$ is not obtained by simply substituting into that nonintegral expansion: a pole in the polylogarithm formula cancels an analytic term and leaves a logarithm.

This article addresses four concrete questions for that marginal endpoint. Does its inverse still have a convergent finite-generator chart? What is the large-order law for the original exponential sectors? Does preserving a coefficient require a cutoff at the fluctuation scale? Can the asymptotic answer be accompanied by a finite, computable error inequality? The answers below are affirmative, explicit, negative, and affirmative, respectively. These are scoped research questions motivated by the inspected corpus, not claims to have settled a named longstanding conjecture in general transseries theory.

Editorial note (ProveIt, 2026-09-29). The logarithmic-endpoint package [5], filed beside this one, treats the same endpoint and model and was written independently of this article, also from a library copy of [4], whose Question 4 both answer at $\alpha = 2$. Its core coincides with this article's: the same Lambert- W_{-1} core (this article's $(w, t, 2r)$, b and A are its (v, T, η) , h and c), the same chart coefficients and inverse diagnostics, the same fold functions, and the same critical cutoff correction, which it states with the weaker error $O(\log \ell_n / \ell_n^{3/2})$. The constants κ_1, κ_2 of Corollary 4.3 are its g_1, g_2 rewritten from its normalization $\ell_n = \log b_n + h$ to $H_n = \log B_n$. It is one theorem with two independent proofs. The explicit contraction radii, the three-generator logarithmic grid, the two-term minimal budget and the finite retention inequality are only here; the Poisson process of large actions, the log-weighted tails, the finite-fold local limit and an exponentiation criterion are only there.

1.2 The main distinction: fluctuations versus the largest action

For the model introduced in Section 2, put

$$N_n = \sqrt{An}, \quad B_n = \sqrt{AnH_n}, \quad H_n = \log B_n.$$

Then $H_n \sim \frac{1}{2} \log n$, so $N_n/B_n \rightarrow 0$. The leading results are

$$[q^n]U_{\beta_*}(q) \sim \frac{\rho^{-n}}{\sqrt{2\pi n B_n}}, \quad (1.1)$$

$$\frac{[q^n]U_{\beta_*}^{[M]}(q)}{[q^n]U_{\beta_*}(q)} \rightarrow \exp\left(-\frac{1}{2\ell^2}\right) \quad \text{if } \frac{M}{N_n} \rightarrow \ell \in (0, \infty). \quad (1.2)$$

The first scale describes the centered sum of actions. The second describes its largest individual actions. At this marginal endpoint the two no longer coincide. In particular, $M \gg B_n$ is sufficient but unnecessarily costly; $M \gg N_n$ is the exact condition for an asymptotically lossless coefficient truncation.

The first correction is also explicit. With $\ell_n = M/N_n$ in a compact subset of $(0, \infty)$,

$$\frac{[q^n]U_{\beta_*}^{[M]}}{[q^n]U_{\beta_*}} = e^{-1/(2\ell_n^2)} \left\{ 1 + \frac{\log(2H_n/\ell_n^2) + 1 - \gamma - 2/\ell_n^2}{4H_n} + O(H_n^{-3/2}) \right\}. \quad (1.3)$$

No finite-prefix constant occurs in this correction. A more general coupling–cutoff version is proved in Theorem 5.2.

1.3 Relation to established work and limits of novelty

The substitution reducing our equation to $z = q \exp(\beta F(z))$ is a standard Lagrange-inversion construction. Its probability interpretation belongs to the established connection between coefficient extraction, random allocation, and conditioned branching structures; Janson gives a broad account, including infinite-variance cases and extremes [8]. The distinction between soft and hard truncation of power tails is also classical [9]. Lambert– W expansions and polylogarithm continuation are standard special-function material [10, 11].

Accordingly, the leading normal-attraction and extreme-value mechanisms are not presented as discoveries. What is developed here is a self-contained, coefficient-level endpoint calculation with convergent local charts, uniform correction terms, cancellation of finite-prefix data, a minimal-budget expansion, and an explicit finite error bound. Establishing whether each combined formula has a previously published equivalent requires a broader literature review than the targeted search performed for this draft.

1.4 Source audit

The repository crosswalk was made against the group

Analysis/Transseries/docs/series-and-transseries/

at the observed revision

3f2d57344dc6f0b3dfa376f9490bfe13d1beb81e.

The group README, the transseries project README, the countable-feedback README, and the descriptions of the moving-fold and adjacent research packages were inspected. The complete approximately seven-hundred-page canonical volume was not audited. The abstract, scope, and research questions of the private stable-tail companion were also inspected. That companion is used to delimit overlap, not as an unproved premise. Every theorem needed below is established here. No repository file was modified.

2 Model, critical geometry, and exact coefficient identities

2.1 Assumptions and notation

Throughout, assume

$$a_j = aj^{-3} + \delta_j \geq 0 \quad (j \geq 1), \quad a > 0, \quad a_1 > 0, \quad \delta_j = 0 \quad (j > J_0), \quad (2.1)$$

for some finite J_0 . The finite perturbations δ_j may have either sign, provided that the actual weights remain nonnegative. Define

$$F(z) = \sum_{j \geq 1} a_j z^j = a \operatorname{Li}_3(z) + P(z), \quad P(z) = \sum_{j \leq J_0} \delta_j z^j, \quad (2.2)$$

$$m = \sum_{j \geq 1} j a_j, \quad \beta_\star = m^{-1}, \quad (2.3)$$

$$A = \beta_\star a, \quad b = \frac{3}{2} + \frac{1}{a} \sum_{j \leq J_0} j^2 \delta_j, \quad (2.4)$$

$$\nu = b + \gamma - \frac{3}{2}, \quad \rho = e^{-\beta_\star F(1)}, \quad U_\star = \beta_\star F(1). \quad (2.5)$$

The second moment $\sum j^2 a_j$ diverges logarithmically, while $F(1)$ and m are finite. We study the small solution of

$$U_\beta(q) = \beta \sum_{j \geq 1} a_j q^j e^{j U_\beta(q)}, \quad U_\beta(0) = 0, \quad (2.6)$$

and its finite-action version $U_\beta^{[M]}$, obtained by replacing F by $F_M(z) = \sum_{j \leq M} a_j z^j$. Write $u_n(\beta) = [q^n]U_\beta$ and $u_n^{[M]}(\beta) = [q^n]U_\beta^{[M]}$.

Substituting $q = e^{-X}$ turns the convergent small- q expansion into an integer-action exponential transseries in X . The critical inverse studied later uses a different variable, $\epsilon = \log(\rho/q)$. The distinction between these two expansions is essential.

Proposition 2.1 (Exact reduction and coefficient identity). *For every $\beta > 0$, equation (2.6) has a unique formal solution with zero constant term, and an analytic solution near zero. Set $z = qe^{U_\beta(q)}$. Then*

$$U_\beta = \beta F(z), \quad q = ze^{-\beta F(z)}, \quad u_n(\beta) = \frac{1}{n} [z^n] e^{n\beta F(z)} \quad (n \geq 1). \quad (2.7)$$

The same statements hold with F_M . All coefficients are nonnegative, $u_n^{[M]}$ increases with M , and $u_n^{[M]} = u_n$ whenever $M \geq n$.

Proof. Each coefficient of the right side of (2.6) depends only on strictly earlier coefficients of U , proving formal existence and uniqueness recursively. Analytic existence follows from the implicit function theorem at $(q, U) = (0, 0)$. The substitution gives $z = q \exp(\beta F(z))$. Lagrange inversion yields

$$[q^n] \beta F(z(q)) = \frac{\beta}{n} [z^{n-1}] F'(z) e^{n\beta F(z)}.$$

Differentiating the exponential shows that the right side equals $n^{-1} [z^n] e^{n\beta F(z)}$. Positivity, monotonicity, and agreement through order M follow directly from this exponential coefficient formula. $\square$

Proposition 2.2 (Poisson representation). *Let Z_j be independent Poisson variables of means $n\beta a_j$, and set $S_n = \sum_{j \geq 1} j Z_j$. This sum is finite almost surely. Then*

$$u_n(\beta) = \frac{e^{n\beta F(1)}}{n} \mathbb{P}(S_n = n). \quad (2.8)$$

If $S_{n,M} = \sum_{j \leq M} j Z_j$, and $\Lambda_{n,M} = n\beta \sum_{j > M} a_j$, then

$$\frac{u_n^{[M]}(\beta)}{u_n(\beta)} = e^{-\Lambda_{n,M}} \frac{\mathbb{P}(S_{n,M} = n)}{\mathbb{P}(S_n = n)}. \quad (2.9)$$

Proof. The total mean number of Poisson events is $n\beta F(1) < \infty$, so only finitely many events occur almost surely. The probability generating function of their total weight is $\exp(n\beta(F(z) - F(1)))$. Extract its n th coefficient and use Proposition 2.1. Independence between the retained and omitted events gives (2.9). $\square$

Proposition 2.3 (Critical boundary). *At $\beta = \beta_*$, the real map $q(z) = ze^{-\beta_* F(z)}$ is strictly increasing on $(0, 1)$ and maps that interval onto $(0, \rho)$. The series for U_{β_*} has radius of convergence exactly ρ , and its continuous boundary value is U_* .*

Proof. For $0 < z < 1$,

$$q'(z) = e^{-\beta_* F(z)}(1 - \beta_* z F'(z)) > 0,$$

because $zF'(z) < m$. The endpoint value is ρ . The Poisson identity bounds $u_n(\beta_*) \leq \rho^{-n}/n$, giving radius at least ρ . The logarithmic inverse in Theorem 3.2 has unbounded first derivative at this endpoint, so the real branch cannot extend analytically through ρ . Thus the radius is exactly ρ . Continuity gives the stated boundary value. $\square$

3 A convergent Lambert chart at the marginal boundary

3.1 The logarithmic normal form

Put $z = e^{-\tau}$, $\tau > 0$, and $\epsilon = \log(\rho/q)$. The exact critical equation becomes

$$\epsilon = \tau + \beta_*(F(e^{-\tau}) - F(1)), \quad U_{\beta_*}(q) = U_* + \epsilon - \tau. \quad (3.1)$$

Lemma 3.1 (Exact analytic remainder). *On the principal logarithmic branch near $\tau = 0$,*

$$\epsilon = \frac{A}{2}\tau^2(\log(1/\tau) + b) + \tau^3 R(\tau), \quad (3.2)$$

where R is holomorphic on $|\tau| < 2\pi$. More explicitly,

$$\tau^3 R(\tau) = \beta_* \sum_{k \geq 3} \frac{(-1)^k}{k!} \left(a\zeta(3-k) + \sum_{j \leq J_0} \delta_j j^k \right) \tau^k. \quad (3.3)$$

For the unperturbed tail, this is $A(\tau^3/12 - \tau^4/288 + \tau^6/86400 + \dots)$.

Proof. The standard integer polylogarithm expansion is

$$\text{Li}_3(e^{-\tau}) = \zeta(3) - \zeta(2)\tau + \frac{\tau^2}{2} \left(\frac{3}{2} - \log \tau \right) + \sum_{k \geq 3} \frac{(-1)^k \zeta(3-k)}{k!} \tau^k.$$

For completeness, subtract the displayed constant, linear, and logarithmic terms. The third derivative of the remainder is

$$-\frac{1}{e^\tau - 1} + \frac{1}{\tau},$$

which is holomorphic for $|\tau| < 2\pi$, including zero. Its Taylor series integrates three times to the indicated remainder. The constant and linear terms in (3.1) cancel because $\beta_* m = 1$; the finite polynomial $P(e^{-\tau})$ supplies the stated quadratic and higher coefficients. This also proves convergence without a formal limiting argument in the polylogarithm order. See [10] for the classical continuation formulas. $\square$

Let $c = A/2$. Define the exact core by

$$w = -W_{-1}\left(-\frac{2\epsilon}{ce^{2b}}\right), \quad t = e^{b-w/2} = \sqrt{\frac{2\epsilon}{cw}}, \quad r = w^{-1}. \quad (3.4)$$

For sufficiently small positive ϵ , $w > 1$, $t \rightarrow 0$, and $r \rightarrow 0$. The core solves $\epsilon = ct^2 \log(e^b/t)$. The negative real branch W_{-1} , not W_0 , is required to obtain a root tending to zero.

Theorem 3.2 (Convergent marginal inverse chart). *There is a function $v(t, r)$, holomorphic near $(0, 0)$, such that*

$$\tau = t(1 + v(t, r)), \quad v(t, 0) = v(0, r) = 0. \quad (3.5)$$

Evaluating (t, r) from (3.4) gives the unique small positive solution of (3.2). Its expansion

$$v(t, r) = \sum_{j \geq 1} V_j(r) t^j \quad (3.6)$$

converges normally on smaller polydiscs. Each V_j is a rational function of r , with a possible pole only at $r = 1$, of order at most $2j - 1$. Its coefficients depend on only the first j Taylor coefficients of R/c . In particular,

$$V_1(r) = -\frac{R(0)}{c} \frac{r}{1-r}.$$

There is no convergent Puiseux representation of τ in any finite ramification $\epsilon^{1/d}$.

Proof. Set $h = 1 + v$. Subtract the exact core equation from (3.2), divide by $ct^2w/2$, and obtain

$$h^2 - 1 - 2rh^2 \log h + \frac{2r}{c} th^3 R(th) = 0. \quad (3.7)$$

This is holomorphic near $(t, r, h) = (0, 0, 1)$, and its derivative with respect to h there is 2. The analytic implicit function theorem gives the unique chart. At $t = 0$ or $r = 0$, the nearby solution is $h = 1$, proving the two vanishing statements.

For the coefficient assertion, write $R(\tau)/c = \sum_{k \geq 0} d_k \tau^k$. At coefficient t^j , the unknown V_j occurs linearly with coefficient $2(1 - r)$. Every other term is a polynomial in $r, d_0, \dots, d_{j-1}$ and earlier V_i . In a product of $k \geq 2$ earlier coefficients whose indices sum to j , the inductive denominator order is at most $2j - k$. Division by $1 - r$ therefore gives at most $2j - 1$. Terms carrying an explicit positive power of t have no larger denominator order. This proves the rationality and pole bound by induction.

For positive τ sufficiently small, the derivative of the right side of (3.2) is

$$c\tau(2 \log(1/\tau) + 2b - 1) + O(\tau^2) > 0.$$

Hence the chart agrees with the unique real inverse. Moreover,

$$\tau \sim \frac{2\sqrt{\epsilon}}{\sqrt{A \log(1/\epsilon)}}. \quad (3.8)$$

Any nonzero convergent Puiseux series has a leading term $C\epsilon^\alpha$, $C \neq 0$. The logarithmic growth ratio in (3.8) forces $\alpha = 1/2$, whereas $\tau/\sqrt{\epsilon} \rightarrow 0$, a contradiction. $\square$

3.2 An explicit holomorphic majorant

The existence assertion can be made quantitative without locating a complex critical value. Choose $R_0 < 2\pi$, and suppose

$$|R(z)| \leq K_0, \quad |R'(z)| \leq K_1 \quad (|z| \leq R_0).$$

Choose $T > 0$ such that

$$\frac{5}{4}T < R_0, \quad \frac{TK_0}{c} \leq \frac{1}{8}, \quad \frac{T^2K_1}{c} \leq \frac{1}{8}. \quad (3.9)$$

Proposition 3.3 (Geometric remainder certificate). *On $|t| \leq T$, $|r| \leq 1/8$, the chart may be chosen with $|v| \leq 1/4$. For fixed $|r| \leq 1/8$ and $|t| < T$, truncating (3.6) after N terms gives*

$$\left| \tau - t \left(1 + \sum_{j=1}^N V_j(r) t^j \right) \right| \leq \frac{|t|}{4} \frac{(|t|/T)^{N+1}}{1 - |t|/T}. \quad (3.10)$$

The Taylor expansion in both t and r has support in $\{(j, k) : j, k \geq 1\}$ for v .

Proof. Equation (3.7) is equivalent to the fixed-point equation

$$v = -\frac{v^2}{2} + r(1+v)^2 \log(1+v) - \frac{rt}{c}(1+v)^3 R(t(1+v)).$$

For $|v| \leq 1/4$, $|\log(1+v)| \leq 1/3$. Under (3.9), the right side has modulus less than $1/4$. Its derivative with respect to v is bounded by

$$\frac{1}{4} + \frac{1}{8} \frac{5}{4} \frac{5}{3} + \frac{1}{8} \left(\frac{3(5/4)^2}{8} + \frac{(5/4)^3}{8} \right) < \frac{5}{8}.$$

Thus it is a uniform contraction on the closed disc. Iteration produces a holomorphic fixed point, and Cauchy's estimate in t gives $|V_j(r)| \leq \frac{1}{4} T^{-j}$. Summing the geometric tail proves the bound. Vanishing on both coordinate axes gives the support assertion. $\square$

Example 3.4 (The pure trilogarithm). For $a_j = j^{-3}$, the first terms, expressed in the exact w -coordinate, are

$$\begin{aligned} \tau = t &- \frac{t^2}{6(w-1)} + \frac{w^2 + 8w - 5}{144(w-1)^3} t^3 \\ &- \frac{9w^3 + 24w^2 - 31w + 10}{1296(w-1)^5} t^4 + O(t^5). \end{aligned} \quad (3.11)$$

The remainder is uniform when $r = 1/w$ stays in the polydisc above. If instead a_1 is increased by $1/2$, the cubic coefficient of $\tau^3 R(\tau)$ vanishes exactly. Thus the first correction to the exact Lambert core disappears. Positivity does not prohibit such local cancellations; the general chart does not assume $R(0) \neq 0$.

3.3 A finite logarithmic generating grid

The exact core can itself be expanded convergently in familiar logarithmic generators. Set

$$L = \log \frac{Ae^{2b}}{4\epsilon}, \quad s = \frac{2\sqrt{\epsilon}}{\sqrt{AL}}, \quad p = \frac{\log L}{L}, \quad u = \frac{1}{L}. \quad (3.12)$$

Corollary 3.5 (Convergent three-generator representation). *There is a holomorphic function $\mathcal{A}(s, p, u)$ near zero with $\mathcal{A}(0, 0, 0) = 1$ such that*

$$\tau = s\mathcal{A}(s, p, u). \quad (3.13)$$

Consequently the critical inverse has an absolutely convergent expansion in three explicit small generators, although it has no finite Puiseux chart. Its first logarithmic block is

$$\tau = s \left[1 - \frac{\log L}{2L} + \frac{3(\log L)^2 - 4\log L}{8L^2} + O\left(\frac{(\log L)^3}{L^3}\right) + O\left(\frac{s}{L}\right) \right]. \quad (3.14)$$

Proof. The core satisfies $w - \log w = L$. Write $w = L(1 + Z)$; then $Z = p + u \log(1 + Z)$. The implicit function theorem gives a holomorphic $Z(p, u)$ with $Z(0, 0) = 0$. Now

$$t = s(1 + Z)^{-1/2}, \quad r = \frac{u}{1 + Z},$$

and substitution in (3.5) proves (3.13). Expanding $Z = p + up + \dots$ gives (3.14). $\square$

The support claim here is deliberately local and explicit. It concerns a convergent analytic series in the stated generators, not a construction of the full field of arbitrary transseries. Conversely, impossibility of a pure-power Puiseux representation is not an obstruction to this logarithmic grid.

4 All-orders Gaussian–logarithmic sector asymptotics

4.1 The marginal normal scale and coupling window

For all sufficiently large n , define the large solution

$$H_n = -\frac{1}{2}W_{-1}\left(-\frac{2}{An}\right), \quad B_n = e^{H_n}, \quad N_n = \sqrt{An}. \quad (4.1)$$

Thus $B_n^2 = AnH_n$, $H_n = \log B_n > 1/2$, and $B_n = N_n\sqrt{H_n}$. The choice of the large branch is part of the normalization. Below, H, B, N abbreviate these three quantities. For a real parameter λ , set

$$\beta_n = \beta_\star \left(1 + \lambda \frac{B_n}{n}\right), \quad \rho_n = e^{-\beta_n F(1)}. \quad (4.2)$$

We always restrict λ to a fixed compact set, so $\beta_n > 0$ eventually. Importantly, ρ_n is a boundary normalization; for $\beta_n > \beta_\star$ it is not asserted to be the radius of convergence of U_{β_n} . The probability identity is valid regardless of that issue.

Use the principal logarithm and put

$$\Psi_b(t) = \frac{t^2}{2}(\log(-it) - b), \quad \Psi_b(0) = 0, \quad \varphi(\lambda) = \frac{e^{-\lambda^2/2}}{\sqrt{2\pi}}. \quad (4.3)$$

For real nonzero t , $\log(-it) = \log|t| - i\pi \operatorname{sgn}(t)/2$. Define the absolutely convergent integrals

$$G_k^{(b)}(\lambda) = \frac{1}{2\pi k!} \int_{\mathbb{R}} e^{i\lambda t - t^2/2} \Psi_b(t)^k dt \quad (k \geq 0). \quad (4.4)$$

Conjugate symmetry makes them real, and $G_0^{(b)} = \varphi$.

Theorem 4.1 (Uniform all-orders sector expansion). *For each fixed integer $K \geq 0$ and compact real λ -interval,*

$$nB_n \rho_n^n u_n(\beta_n) = \sum_{k=0}^K \frac{G_k^{(b)}(\lambda)}{H_n^k} + O(H_n^{-K-1}), \quad (4.5)$$

uniformly on that interval. The constants may depend on the fixed weights, K , and the interval. This is an asymptotic expansion; convergence of the infinite series in $1/H_n$ is not part of the assertion.

We prove the Fourier estimates rather than assuming a local limit theorem. This also supplies the estimates required for the moving cutoff.

Lemma 4.2 (Fourier damping from the exact tail). *Let*

$$D(\theta) = \sum_{j \geq 1} a_j (1 - \cos j\theta), \quad D_M(\theta) = \sum_{j \leq M} a_j (1 - \cos j\theta).$$

There are positive constants c_1, c_2 and $0 < \theta_0 < 1$ such that

$$D(\theta) \geq c_1 \theta^2 \log(1/|\theta|) \quad (0 < |\theta| \leq \theta_0), \quad D(\theta) \geq c_2 \quad (\theta_0 \leq |\theta| \leq \pi).$$

If M/N_n lies in a fixed compact subset of $(0, \infty)$, the following bounds hold, with possibly smaller constants, both for D and D_M :

$$\begin{aligned} n\beta_n D(t/B) &\geq c_1 t^2 & (|t| \leq \sqrt{B}), \\ n\beta_n D(t/B) &\geq c_1 B & (\sqrt{B} \leq |t| \leq \theta_0 B), \\ n\beta_n D(t/B) &\geq c_2 n & (\theta_0 B \leq |t| \leq \pi B). \end{aligned}$$

The displayed statements for D_M mean that D is replaced by D_M .

Proof. For $|j\theta| \leq 1$, $1 - \cos(j\theta) \geq c(j\theta)^2$. Summing the exact tail for $J_0 < j \leq 1/|\theta|$ gives the logarithm. For a cutoff, sum instead up to $\min(M, 1/|\theta|)$. Bounded initial terms can be absorbed by taking θ_0 small; the bound needed is $c\theta^2 \log \min(M, 1/|\theta|)$ when that minimum is large. For $|t| \leq \sqrt{B}$, both M and $B/|t|$ are at least a fixed multiple of $\sqrt{B}$, eventually. Their minimum has logarithm at least $H/3$, and $n/B^2 = 1/(AH)$. For $\sqrt{B} \leq |t| \leq \theta_0 B$, we have $1/|\theta| \leq \sqrt{B} < M$. Choose θ_0 small enough that $\theta^2 \log(1/\theta)$ increases on $(0, \theta_0]$. Its value at $B^{-1/2}$, after multiplication by n , is of order B . Finally, $a_1(1 - \cos \theta)$ is bounded below on $[\theta_0, \pi]$. All constants remain uniform when β_n remains in a compact positive neighborhood of $\beta_\star$. $\square$

Proof of Theorem 4.1. Fourier inversion and Proposition 2.2 give

$$nB \rho_n^n u_n(\beta_n) = \frac{1}{2\pi} \int_{-\pi B}^{\pi B} \exp \left\{ n\beta_n (F(e^{it/B}) - F(1)) - \frac{int}{B} \right\} dt. \quad (4.6)$$

The boundary values of the expansion in Lemma 3.1 give, near zero,

$$\beta_\star (F(e^{-s}) - F(1)) = -s + \frac{A}{2} s^2 (\log(1/s) + b) + s^3 R(s).$$

The identity remains valid on either imaginary side, with the specified logarithm. Substitute $s = -it/B$. The linear term, after centering at n , is exactly $i\lambda t$; the quadratic logarithm is $-t^2/2 + \Psi_b(t)/H$. Therefore, for $|t| \leq H^{1/8}$, the exponent in (4.6) equals

$$i\lambda t - \frac{t^2}{2} + \frac{\Psi_b(t)}{H} + E_n(t), \quad (4.7)$$

where one can bound $E_n(t)$ by a constant times

$$\frac{B}{n} \left(t^2 + \frac{t^2(1 + |\log |t||)}{H} \right) + \frac{n}{B^3} |t|^3.$$

Here $t^2 |\log |t||$ is assigned its continuous value at zero. Both prefactors are smaller than every fixed inverse power of H , including after multiplication by a fixed power of H .

On this interval $\Psi_b/H = o(1)$ uniformly. Taylor's formula for its exponential, integrated against the Gaussian, has remainder $O(H^{-K-1})$, because $\int e^{-t^2/4} |\Psi_b(t)|^{K+1} dt < \infty$. The same Gaussian majorant absorbs E_n . By Lemma 4.2, the rest of the exact Fourier integral is smaller than every power of $1/H$: first there is a Gaussian tail beyond $H^{1/8}$, then a bound $O(Be^{-cB})$, and finally $O(Be^{-cn})$. The tails of each approximating Gaussian–logarithmic integral are equally negligible. Extending those integrals to $\mathbb{R}$ gives (4.5). Every estimate used is uniform for bounded λ . $\square$

4.2 Explicit constants and a finite formula for every order

Corollary 4.3 (Critical coefficient expansion). *At $\beta = \beta_\star$,*

$$u_n(\beta_\star) = \frac{\rho^{-n}}{\sqrt{2\pi} n B_n} \left[1 + \frac{\kappa_1}{H_n} + \frac{\kappa_2}{H_n^2} + O(H_n^{-3}) \right], \quad (4.8)$$

where

$$\kappa_1 = \frac{2 - \gamma - \log 2 - 2b}{4}, \quad (4.9)$$

$$\kappa_2 = \frac{3}{8} \left[\left(\frac{4}{3} - \frac{\gamma + \log 2}{2} - b \right)^2 - \frac{\pi^2}{8} - \frac{10}{9} \right]. \quad (4.10)$$

In particular,

$$u_n(\beta_\star) \sim \frac{\rho^{-n}}{\sqrt{\pi A} n^{3/2} \sqrt{\log n}}. \quad (4.11)$$

Proof. For a standard normal variable T , the first coefficient is $\frac{1}{2} \mathbb{E}[T^2(\log |T| - b)]$. Differentiate $\mathbb{E}|T|^s = 2^{s/2} \Gamma((s+1)/2) / \sqrt{\pi}$ at $s = 2$ to obtain $\mathbb{E}[T^2 \log |T|] = 1 - (\gamma + \log 2)/2$. At the second order, conjugate symmetry replaces the square of $\log(-it) - b$ by $(\log |t| - b)^2 - \pi^2/4$. Under the probability measure proportional to $t^4 e^{-t^2/2} dt$, $\log |t|$ has mean $4/3 - (\gamma + \log 2)/2$ and variance $\pi^2/8 - 10/9$. Since $\mathbb{E}T^4 = 3$, formula (4.10) follows. Finally, $H_n \sim \frac{1}{2} \log n$ gives (4.11). $\square$

A convenient nonrecursive prescription for every critical coefficient uses

$$\mathcal{M}(\alpha) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-t^2/2} (-it)^\alpha dt = \frac{2^{(\alpha-1)/2}}{\pi} \Gamma\left(\frac{\alpha+1}{2}\right) \cos\left(\frac{\pi\alpha}{2}\right) \quad (\Re \alpha > -1). \quad (4.12)$$

Differentiation under the integral is justified on compact subsets of that half-plane. Since $t^{2k} = (-1)^k (-it)^{2k}$,

$$G_k^{(b)}(0) = \frac{(-1)^k}{2^k k!} (\partial_\alpha - b)^k \mathcal{M}(\alpha) \Big|_{\alpha=2k}. \quad (4.13)$$

Thus each coefficient reduces to finitely many gamma derivatives and ordinary arithmetic. This formula is not an assertion about the growth of these coefficients as $k \rightarrow \infty$.

5 A universal cutoff law and the minimal action budget

5.1 A quantitative local limit for the truncated action sum

In this section M is an integer, $\ell_n = M/N_n$, and ℓ_n stays in a compact subset of $(0, \infty)$. Define

$$\begin{aligned}\Lambda_n &= n\beta_n \sum_{j>M} a_j, \\ \xi_n &= \frac{n - n\beta_n \sum_{j\leq M} ja_j}{B_n}, \quad v_n = \frac{n\beta_n \sum_{j\leq M} j^2 a_j}{B_n^2}.\end{aligned}\tag{5.1}$$

The notation v_n denotes a normalized variance, not the local inverse correction $v(t, r)$ from Section 3.

Lemma 5.1 (Truncated Gaussian local approximation). *Uniformly for bounded λ and the stated ℓ_n -range,*

$$B_n \mathbb{P}(S_{n,M} = n) = v_n^{-1/2} \varphi\left(\frac{\xi_n}{\sqrt{v_n}}\right) + O(H_n^{-3/2}).\tag{5.2}$$

Writing $\ell = \ell_n$ and

$$d_n = \nu + \log \ell - \frac{1}{2} \log H_n,\tag{5.3}$$

we also have

$$\begin{aligned}\Lambda_n &= \frac{1}{2\ell^2} + o(H_n^{-K}), \\ \xi_n &= -\lambda + \frac{1}{\ell\sqrt{H_n}} + O(N_n^{-1}), \\ v_n &= 1 + \frac{d_n}{H_n} + o(H_n^{-K})\end{aligned}\tag{5.4}$$

for each fixed K , with the little-oh statements uniform on the same compact sets.

Proof. For $M \geq J_0$, elementary tail estimates and the harmonic-number expansion give

$$\begin{aligned}\sum_{j>M} a_j &= \frac{a}{2M^2} + O(M^{-3}), \\ \sum_{j>M} ja_j &= \frac{a}{M} + O(M^{-2}), \\ \sum_{j\leq M} j^2 a_j &= a(\log M + \nu) + O(M^{-1}).\end{aligned}\tag{5.5}$$

Also $\sum_{j\leq M} j^3 a_j = O(M)$. The exact centered Fourier exponent for $S_{n,M}$, at frequency t/B , is

$$-i\xi_n t - \frac{v_n t^2}{2} + E_{n,M}(t), \quad |E_{n,M}(t)| \leq \frac{n\beta_n |t|^3}{6B^3} \sum_{j\leq M} j^3 a_j \leq C|t|^3 H^{-3/2}.$$

The elementary inequality used here is $|e^{ix} - 1 - ix + x^2/2| \leq |x|^3/6$ for real x . Expand on $|t| \leq H^{1/8}$ and use the same Gaussian damping and negligible complementary tails as in Theorem 4.1. The integral error is $O(H^{-3/2})$. Fourier inversion of the quadratic exponent gives (5.2). Substitution of $M = \ell N$, $B = N\sqrt{H}$, and $\log N = H - \frac{1}{2} \log H$ in (5.5) proves (5.4). Errors involving $1/N$ or B/n are smaller than every fixed power of $1/H$. $\square$

5.2 Coupling–cutoff correction and cancellation of prefix data

Define a real function by the convergent integral

$$\mathcal{J}(\lambda) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\lambda t - t^2/2} \frac{t^2}{2} \log(-it) \, dt. \quad (5.6)$$

In particular, $\mathcal{J}(0)/\varphi(0) = (2 - \gamma - \log 2)/4$.

Theorem 5.2 (Universal moving-window cutoff correction). *Let $R_n(M; \lambda) = u_n^{[M]}(\beta_n)/u_n(\beta_n)$. For λ in a fixed compact subset of $\mathbb{R}$ and $\ell = \ell_n = M/N_n$ in a fixed compact subset of $(0, \infty)$,*

$$\begin{aligned} R_n(M; \lambda) = e^{-1/(2\ell^2)} & \left[1 + \frac{\lambda}{\ell\sqrt{H}} \right. \\ & \left. + \frac{\lambda^2 - 1}{2H} \left(\gamma - \frac{3}{2} + \log \ell - \frac{1}{2} \log H + \frac{1}{\ell^2} \right) - \frac{\mathcal{J}(\lambda)}{H\varphi(\lambda)} + O\left(\frac{\log H}{H^{3/2}}\right) \right]. \end{aligned} \quad (5.7)$$

In particular, at critical coupling,

$$R_n(M; 0) = e^{-1/(2\ell^2)} \left[1 + \frac{\log(2H/\ell^2) + 1 - \gamma - 2/\ell^2}{4H} + O(H^{-3/2}) \right]. \quad (5.8)$$

Apart from the scale A entering N, H , these displayed corrections are unchanged by every finite perturbation allowed in (2.1).

Proof. By the exact Poisson identity,

$$R_n(M; \lambda) = e^{-\Lambda_n} \frac{\mathbb{P}(S_{n,M} = n)}{\mathbb{P}(S_n = n)}.$$

The full denominator, multiplied by B , is $\varphi(\lambda) + G_1^{(b)}(\lambda)/H + O(H^{-2})$. Its leading term is bounded away from zero on a compact λ -set. Taylor expansion of the Gaussian term in Lemma 5.1 gives

$$\frac{v_n^{-1/2} \varphi(\xi_n/\sqrt{v_n})}{\varphi(\lambda)} = 1 + \frac{\lambda}{\ell\sqrt{H}} + \frac{\lambda^2 - 1}{2H} \left(d_n + \frac{1}{\ell^2} \right) + O\left(\frac{\log H}{H^{3/2}}\right). \quad (5.9)$$

Indeed, the displacement of the mean is $1/(\ell\sqrt{H})$ and the variance displacement is $d_n/H = O(\log H/H)$; retaining the square of the former produces the $1/\ell^2$ term. The mixed term has size $O(\log H/H^{3/2})$, and the square of the variance displacement is smaller than that order.

Differentiating the Gaussian Fourier transform twice shows

$$G_1^{(b)}(\lambda) = \mathcal{J}(\lambda) + \frac{b}{2}(\lambda^2 - 1)\varphi(\lambda). \quad (5.10)$$

Divide (5.9) by the normalized full probability, multiply by $e^{-\Lambda_n}$, and use $\nu - b = \gamma - 3/2$. This cancels b exactly and proves (5.7).

When $\lambda = 0$, odd first-order mean terms and the mixed mean–variance term vanish. The remaining Taylor error is $O((\log H)^2/H^2) = o(H^{-3/2})$, in addition to the local-limit error $O(H^{-3/2})$. Finally,

$$-\frac{d_n}{2} - \frac{1}{2\ell^2} - \kappa_1 = \frac{\log(2H/\ell^2) + 1 - \gamma - 2/\ell^2}{4},$$

which proves (5.8) with its improved error. $\square$

Remark 5.3 (What the universality statement does not say). Finite-prefix changes alter A, b, ρ , the exact inverse chart, and the full-sector coefficients κ_k . They do not disappear from the entire problem. The theorem identifies the cancellation in the specific normalized cutoff ratio through the orders displayed. It makes no all-orders prefix-independence claim.

5.3 Sharp scale, necessity, and an optimally selected budget

Corollary 5.4 (Complete cutoff trichotomy). *At critical coupling,*

$$R_n(M; 0) \longrightarrow \begin{cases} 0, & M/N_n \longrightarrow 0, \\ e^{-1/(2\ell^2)}, & M/N_n \longrightarrow \ell \in (0, \infty), \\ 1, & M/N_n \longrightarrow \infty. \end{cases} \quad (5.11)$$

Moreover, $R_n(M; 0) \rightarrow 1$ if and only if $M/N_n \rightarrow \infty$. The same conclusions hold in the moving window for any bounded sequence $\lambda = \lambda_n$.

Proof. For finite positive ℓ , apply Theorem 5.2. For $M/N \rightarrow 0$, monotonicity in M bounds the limsup by $e^{-1/(2\varepsilon^2)}$ for each fixed $\varepsilon > 0$; let $\varepsilon \downarrow 0$. For $M/N \rightarrow \infty$, the liminf is at least $e^{-1/(2L^2)}$ for each fixed $L > 0$; let $L \rightarrow \infty$ and use $R_n \leq 1$. If M/N fails to tend to infinity, a subsequence is bounded above by a fixed L , and its ratio has limsup at most $e^{-1/(2L^2)} < 1$. Uniformity in bounded λ gives the last assertion. Integer parts in the comparison cutoffs have no effect. $\square$

Theorem 5.5 (Two-term minimal action budget). *Fix $0 < \eta < 1$, let $d_\eta = -\log(1 - \eta)$, and define*

$$M_{\min}(n, \eta) = \min\{M \geq 1 : R_n(M; 0) \geq 1 - \eta\}.$$

Then

$$M_{\min}(n, \eta) = \frac{N_n}{\sqrt{2d_\eta}} \left[1 - \frac{\log(4d_\eta H_n) + 1 - \gamma - 4d_\eta}{8d_\eta H_n} + O(H_n^{-3/2}) \right]. \quad (5.12)$$

The error statement is for fixed η ; it is also uniform when η stays in a compact subset of $(0, 1)$.

Proof. Existence follows from exact equality of the coefficients for $M \geq n$. The leading profile and monotonicity first imply $M_{\min}/N \rightarrow \ell_0 = (2d_\eta)^{-1/2}$. Take logarithms in (5.8). The square of the correction is $o(H^{-3/2})$, so

$$\log R_n(M; 0) = -\frac{1}{2\ell^2} + \frac{C(\ell, H)}{H} + O(H^{-3/2}), \quad C(\ell, H) = \frac{\log(2H/\ell^2) + 1 - \gamma - 2/\ell^2}{4}.$$

Writing $\ell = \ell_0(1 + z/H)$, the first term is $-d_\eta + 2d_\eta z/H + O(z^2/H^2)$. The threshold therefore occurs at

$$z = -\frac{\log(4d_\eta H) + 1 - \gamma - 4d_\eta}{8d_\eta},$$

up to an error $O(H^{-1/2})$. More explicitly, perturbing the proposed ℓ by $\pm C_0 H^{-3/2}$, with C_0 sufficiently large, gives opposite strict inequalities for the threshold because the derivative of $-1/(2\ell^2)$ is bounded below near ℓ_0 . Monotonicity traps the first crossing. Rounding to the next integer changes ℓ by $O(N^{-1}) = o(H^{-K})$ for every fixed K . $\square$

The theorem supplies an asymptotic optimizer, not an unconditional finite error guarantee obtained by discarding its big-oh term. The next section provides such a guarantee separately.

6 A finite error certificate and proof-producing computation

6.1 An elementary anti-concentration bound

The following estimate needs less than the exact-tail assumption. Suppose only that $a_j \geq 0$, $a_1 > 0$, $\sum a_j < \infty$, and $\beta > 0$. The finite sums below are always defined. Let S_n be the compound Poisson sum of Proposition 2.2.

Proposition 6.1 (Explicit bound on every point probability). *For any integer $J \geq 1$ and $0 < \vartheta \leq 1$, put*

$$V_J = \sum_{j \leq J} j^2 a_j, \quad I_{n,J,\vartheta} = \frac{1}{\sqrt{2\pi n\beta(1 - \vartheta^2/12)V_J}} + \exp\left(-\frac{2n\beta a_1 \vartheta^2}{\pi^2 J^2}\right). \quad (6.1)$$

Then $\sup_{k \in \mathbb{Z}} \mathbb{P}(S_n = k) \leq I_{n,J,\vartheta}$.

Proof. Taking absolute values in Fourier inversion gives

$$\mathbb{P}(S_n = k) \leq \frac{1}{\pi} \int_0^\pi \exp\left\{-n\beta \sum_{j \geq 1} a_j (1 - \cos j\theta)\right\} d\theta.$$

If $0 \leq \theta \leq \vartheta/J$, then

$$1 - \cos(j\theta) \geq \frac{j^2 \theta^2}{2} \left(1 - \frac{\vartheta^2}{12}\right) \quad (j \leq J).$$

Extend the resulting Gaussian integral to infinity to obtain the first term of (6.1). On the rest of $[0, \pi]$, the single weight a_1 gives $\sum a_j (1 - \cos j\theta) \geq a_1 (1 - \cos \theta) \geq 2a_1 \theta^2 / \pi^2$. Bounding the integrand by its value at ϑ/J and the interval length by π gives the second term. The estimates are independent of k . $\square$

6.2 A positive coefficient recurrence with a certified tail inequality

For fixed n, M, β , define

$$P_{n,M} = e^{-n\beta F(1)} [z^n] e^{n\beta F_M(z)}, \quad \Lambda = n\beta \sum_{j > M} a_j. \quad (6.2)$$

This normalization uses the full $F(1)$, not $F_M(1)$. Accordingly, $P_{n,M}$ is the probability of $S_n = n$ and no omitted action occurring.

Theorem 6.2 (Finite coefficient-retention certificate). *For every $n, M, J \geq 1$, $\beta > 0$, and $0 < \vartheta \leq 1$,*

$$\boxed{\frac{P_{n,M}}{P_{n,M} + \Lambda I_{n,J,\vartheta}} \leq \frac{u_n^{[M]}(\beta)}{u_n(\beta)} \leq 1.} \quad (6.3)$$

All quantities on the left are explicit. In the exact-tail model at critical coupling and $M \geq J_0$, one may replace Λ by the upper bound $nA/(2M^2)$.

Proof. Let $Z_{>M} = \sum_{j>M} Z_j$. The missed probability satisfies

$$\begin{aligned} \mathbb{P}(S_n = n, Z_{>M} \geq 1) &\leq \mathbb{E}[Z_{>M} \mathbf{1}_{\{S_n=n\}}] \\ &= \sum_{j>M} n\beta a_j \mathbb{P}(S_n = n - j) \\ &\leq \Lambda I_{n,J,\vartheta}. \end{aligned}$$

The middle equality is the elementary Poisson shift identity $\mathbb{E}[Z_j h(S_n)] = n\beta a_j \mathbb{E}[h(S_n + j)]$, proved by shifting the index in the Poisson mass function. Nonnegative terms justify summing over the countable family. In fact, terms with $j > n$ vanish, though using the whole tail is a convenient upper bound. Since

$$\mathbb{P}(S_n = n) = P_{n,M} + \mathbb{P}(S_n = n, Z_{>M} \geq 1),$$

and the desired coefficient ratio is $P_{n,M}/\mathbb{P}(S_n = n)$, the result follows. Finally, $\sum_{j>M} j^{-3} \leq \int_M^\infty x^{-3} dx = 1/(2M^2)$. $\square$

The coefficient $P_{n,M}$ is computable by a recurrence containing no subtractions. With $p_0 = e^{-n\beta F(1)}$, define

$$p_k = \frac{n\beta}{k} \sum_{j=1}^{\min(k,M)} j a_j p_{k-j} \quad (1 \leq k \leq n). \quad (6.4)$$

Then $p_n = P_{n,M}$. This follows by differentiating $\sum p_k z^k = \exp(n\beta(F_M(z) - F(1)))$. The arithmetic-operation count is $O(nM)$, and an implementation needs only the preceding M values, hence $O(M)$ working storage. These are arithmetic-operation bounds, not bit-complexity or rounding-error claims. The reference script keeps all coefficients for simplicity.

For a literal machine certificate, evaluate a lower enclosure $P_- \leq P$ and upper enclosures $\Lambda_+ \geq \Lambda$, $I_+ \geq I$, all nonnegative. Then

$$\frac{P_-}{P_- + \Lambda_+ I_+}$$

is a valid lower retention bound, with its final division also rounded downward. The mpmath diagnostics supplied here do not perform that directed rounding. Thus Theorem 6.2 is a proved mathematical certificate, whereas the recorded floating-point values illustrate it rather than instantiate a proof-assistant or interval-arithmetic certificate.

6.3 The finite certificate attains the sharp order of the action budget

The Fourier majorant can be made asymptotically sharp enough to preserve the $\sqrt{n}$ scale. Let

$$J_n = \max\left(1, \left\lfloor \frac{\sqrt{n}}{(\log n)^{3/2}} \right\rfloor\right), \quad \vartheta_n = (\log n)^{-1/2}, \quad (6.5)$$

for n sufficiently large that $\vartheta_n \leq 1$.

Corollary 6.3 (Asymptotic cost of the finite certificate). *At critical coupling,*

$$I_{n,J_n,\vartheta_n} \sim \frac{1}{\sqrt{2\pi} B_n}.$$

If $M/N_n \rightarrow \ell \in (0, \infty)$, and $t = 1/(2\ell^2)$, the lower bound in (6.3) tends to

$$\frac{1}{1 + te^t}. \quad (6.6)$$

Consequently, for a fixed tolerance $0 < \eta < 1$, every fixed constant

$$\ell > \frac{1}{\sqrt{2W_0(\eta/(1-\eta))}} \quad (6.7)$$

is sufficient for the finite inequality eventually to certify $R_n(\lceil \ell N_n \rceil; 0) \geq 1 - \eta$. The ratio of this threshold constant to the optimal asymptotic constant is

$$\sqrt{\frac{-\log(1-\eta)}{W_0(\eta/(1-\eta))}} \rightarrow 1 \quad (\eta \downarrow 0). \quad (6.8)$$

The limits here are first $n \rightarrow \infty$ for fixed η , then $\eta \downarrow 0$.

Proof. We have $V_{J_n} \sim a \log J_n \sim a H_n$, so the Gaussian term in (6.1) is asymptotic to $1/(\sqrt{2\pi}B_n)$. The second term is $\exp(-c(\log n)^2(1+o(1))) = o(B_n^{-1})$. Theorem 5.2 and the local limit give $P_{n,M} \sim e^{-t}/(\sqrt{2\pi}B_n)$, while $\Lambda \rightarrow t$. Substitution proves (6.6). Its value exceeds $1 - \eta$ precisely when $te^t < \eta/(1 - \eta)$, proving (6.7). Finally, both $-\log(1 - \eta)$ and $W_0(\eta/(1 - \eta))$ are asymptotic to η . $\square$

This comparison does not make the elementary majorant optimal at every fixed tolerance. It proves that an actual finite inequality, rather than an uncontrolled asymptotic substitution, operates on the correct action scale. For a prescribed finite n , one evaluates the inequality itself.

7 How the finite-action folds approach the marginal boundary

7.1 A second convergent logarithmic chart

Fix $\beta = \beta_\star$. The finite map $q = ze^{-\beta_\star F_M(z)}$ has a unique positive critical point $z_M = e^{v_M} > 1$, characterized by

$$\beta_\star \sum_{j \leq M} j a_j e^{j v_M} = 1. \quad (7.1)$$

Indeed, the left side is strictly increasing in v_M , is below one at zero, and tends to infinity. Put

$$L_M = \log M + \nu, \quad \rho_M = e^{v_M - \beta_\star F_M(e^{v_M})}.$$

The reuse of the letter L with subscript M should not be confused with the inverse variable L in (3.12). Define entire functions

$$\begin{aligned} K(s) &= \sum_{k \geq 2} \frac{s^k}{k!(k-1)} = \int_0^1 \frac{e^{sx} - 1 - sx}{x^2} dx, \\ Q(s) &= \sum_{k \geq 3} \frac{s^k}{k!(k-2)} = \int_0^1 \frac{e^{sx} - 1 - sx - (sx)^2/2}{x^3} dx. \end{aligned} \quad (7.2)$$

The integrands have removable values at zero.

Theorem 7.1 (Logarithmic fold chart, drift, and curvature). *There is a holomorphic $S(r)$ near zero, uniquely determined by*

$$S + rK(S) = r, \quad S(0) = 0. \quad (7.3)$$

It has a convergent Taylor series, beginning

$$S(r) = r - \frac{1}{2}r^3 - \frac{1}{12}r^4 + O(r^5). \quad (7.4)$$

For the finite-action critical point,

$$v_M = \frac{1}{M}S(1/L_M) + O\left(\frac{1}{M^2L_M}\right). \quad (7.5)$$

Its critical value satisfies

$$\log \frac{\rho_M}{\rho} = \frac{A}{M^2} \left[\frac{1}{2} + S - \frac{L_M}{2}S^2 - Q(S) \right] + O(M^{-3}), \quad S = S(1/L_M), \quad (7.6)$$

and hence

$$\log \frac{\rho_M}{\rho} = \frac{A}{2M^2} \left[1 + \frac{1}{L_M} + O(L_M^{-3}) \right] + O(M^{-3}). \quad (7.7)$$

Finally, the positive curvature is

$$\kappa_M := \beta_\star \sum_{j \leq M} j^2 a_j e^{jv_M} = A [L_M + K'(S(1/L_M))] + O(M^{-1}). \quad (7.8)$$

Proof. The implicit function theorem applied to $S + rK(S) - r$ at zero gives $S(r)$, since the derivative with respect to S is one. Coefficient comparison gives (7.4).

Write $s = Mv_M$. First observe that $s = O(1/L_M)$: by positivity, $e^{jv} - 1 \geq jv$, so the critical equation implies

$$v_M \sum_{j \leq M} j^2 a_j \leq \sum_{j > M} j a_j.$$

The moment estimates in (5.5) give the assertion. Expand the left side of (7.1) after subtracting its value at zero. The linear term is $aL_M s/M + O(M^{-2})$. For the exact tail, the nonlinear part, multiplied by M/a , is

$$\frac{1}{M} \sum_{j \leq M} \frac{e^{s(j/M)} - 1 - s(j/M)}{(j/M)^2} = K(s) + O(M^{-1}),$$

uniformly for s in a bounded complex neighborhood of zero. This is a Riemann sum for a function analytic jointly in s, x after removing its value at $x = 0$; its first x -derivative is uniformly bounded on the relevant compact set. The finitely many perturbations contribute $O(M^{-1})$ after the same scaling. The omitted first moment is $a/M + O(M^{-2})$. We obtain

$$L_M s + K(s) = 1 + O(M^{-1}). \quad (7.9)$$

The derivative on the left is $L_M + O(1)$. Comparison with the exact solution $S(1/L_M)$ gives $s - S = O(1/(ML_M))$, proving (7.5).

For the critical-value drift, Taylor expansion at $z = 1$ in the logarithmic coordinate gives

$$\begin{aligned} \log \frac{\rho_M}{\rho} &= \beta_\star (F(1) - F_M(1)) + v_M \left(1 - \beta_\star \sum_{j \leq M} j a_j \right) \\ &\quad - \frac{\beta_\star v_M^2}{2} \sum_{j \leq M} j^2 a_j - \beta_\star \sum_{j \leq M} a_j \left(e^{jv_M} - 1 - jv_M - \frac{(jv_M)^2}{2} \right). \end{aligned}$$

The four terms, multiplied by M^2/A , are respectively $1/2 + O(M^{-1})$, $s + O(M^{-1})$, $-L_M s^2/2 + O(M^{-1})$, and $-Q(s) + O(M^{-1})$. The last estimate follows from the analogous analytic Riemann

sum with an x^{-3} denominator. Replacing s by S changes this scaled expression by at most $O(M^{-1})$, proving (7.6). To expand it, use $L_M S = 1 - K(S)$; the bracket equals

$$\frac{1}{2} + \frac{S}{2} + \frac{SK(S)}{2} - Q(S) = \frac{1}{2} + \frac{1}{2L_M} + O(L_M^{-3}).$$

This also explains the absence of a L_M^{-2} term.

Finally,

$$\sum_{j \leq M} j^2 a_j (e^{jv_M} - 1) = a \int_0^1 \frac{e^{sx} - 1}{x} dx + O(M^{-1}) = aK'(s) + O(M^{-1}),$$

by a third analytic Riemann-sum estimate; finite-prefix changes are $O(v_M)$. Add the unperturbed truncated second moment and replace s by S . This proves (7.8). $\square$

7.2 The analytic fold law and why the two limits differ

For each fixed M , put $U_M = \beta_\star F_M(e^{v_M})$. On the positive small-solution sheet, as $q \uparrow \rho_M$,

$$U_{\beta_\star}^{[M]}(q) = U_M - \sqrt{\frac{2 \log(\rho_M/q)}{\kappa_M}} + O(\log(\rho_M/q)). \quad (7.10)$$

To see this directly, write $v = \log z$. The derivative of $\log q = v - \beta_\star F_M(e^v)$ vanishes at v_M , and its second derivative there is $-\kappa_M$. The derivative of $\beta_\star F_M(e^v)$ at the same point is one. The analytic square-root inversion of this nondegenerate quadratic zero gives (7.10).

Every finite model therefore has an ordinary analytic fold. Nevertheless, $\kappa_M \sim A \log M$ diverges, $v_M \sim 1/(M \log M)$, and the limiting boundary has the logarithmic inverse of Section 3. At a cutoff $M \sim \ell N_n$, the critical-value drift satisfies

$$n \log(\rho_M/\rho) \longrightarrow \frac{1}{2\ell^2}. \quad (7.11)$$

Its exponential is exactly the leading suppression factor in the cutoff law. This is an explanatory compatibility check, not a substitute for a uniform coefficient theorem: fixed- M singularity analysis alone would not justify letting M grow with n . Theorem 5.2 was proved independently by uniform Fourier estimates.

8 Reproducible verification and numerical examples

8.1 What was checked exactly

The package contains two executable Python programs and their recorded JSON outputs. The main program performs 102 exact checks: 96 comparisons of a rational exponential-series recurrence with independent partition enumeration; four coefficients of the reduced Lambert-core equation; the cubic cancellation caused by increasing a_1 by $1/2$; and the algebraic cancellation of b in the critical cutoff correction. The partition tests use $n = 1, \dots, 12$, $\beta = 1/3, 2/3$, two prefixes, and both a fixed small cutoff and the full degree cutoff. They check the coefficient before the common factor $1/n$.

The supplementary program performs nine further exact checks: eight coefficients of the fold equation (7.3), through degree seven, and the finite-prefix cancellation for general λ . Thus 111 finite exact checks passed in the recorded runs. The independent fold computation gives

$$S(r) = r - \frac{r^3}{2} - \frac{r^4}{12} + \frac{35r^5}{72} + \frac{33r^6}{160} - \frac{1013r^7}{1800} + O(r^8).$$

These checks do not verify infinitely many coefficients or replace any analytic proof above.

8.2 Inverse accuracy and sector coefficients

For the pure weights $a_j = j^{-3}$,

$$A = 0.60792710185402662866\dots, \quad b = \frac{3}{2},$$

and the first two normalized sector constants are

$$\kappa_1 = -0.56759071136536954251\dots, \quad \kappa_2 = -0.63819423854137909958\dots$$

Direct quadrature of the Gaussian–logarithmic moments agrees with the closed formulas within 2×10^{-50} in the fifty-digit run. A separate comparison at $n = 256$ checks the Fourier calculation against the positive coefficient recurrence, for both the full coefficient and $M = \lfloor N_n \rfloor$. The relative discrepancies are below 10^{-48} . These are floating-point agreement figures, not rigorous error enclosures.

ϵ	τ , numerical root	Core relative error	Three-term relative error
10^{-4}	$7.146046748 \times 10^{-3}$	1.0020×10^{-4}	2.7144×10^{-11}
10^{-8}	$5.388914675 \times 10^{-5}$	4.1471×10^{-7}	2.9478×10^{-18}
10^{-16}	$3.972838893 \times 10^{-9}$	1.6274×10^{-11}	3.0045×10^{-31}

Table 1: Pure-tail inverse diagnostics. The core is t ; the three-term approximation retains t, t^2, t^3 in (3.11).

n	H_n	$\sqrt{2\pi} n B_n \rho^n u_n$	$1 + \kappa_1/H_n + \kappa_2/H_n^2$
10^4	5.1785861	0.87607723	0.86659913
10^8	10.1186819	0.93893996	0.93767354
10^{16}	19.6611530	0.96965125	0.96948041

Table 2: Pure-tail sector coefficients and their first two corrections. Convergence in the logarithmic variable is slow even at very large n .

The inverse chart is genuinely convergent; the outer coefficient expansion is only asymptotic. The sharp contrast between Tables 1 and 2 is therefore expected. A large value of n need not make $1/H_n$ extremely small.

8.3 Cutoff, prefix, and moving-window diagnostics

The cutoff calculations use the exact integer M , not its unrounded nominal value, in $\ell = M/N_n$. Table 3 uses $M = \lfloor N_n \rfloor$. The complete JSON file additionally contains nominal $\ell = 1/2, 2$, both prefixes, and more digits.

At $n = 10^8$, pure weights, and $M = 7796$, the moving-window ratios are approximately 0.45615931 for $\lambda = -1$ and 0.75444199 for $\lambda = 1$. Formula (5.7) gives respectively 0.49001174 and 0.74751311, versus a common leading value 0.60645535. These values illustrate the sizable $H^{-1/2}$ coupling correction. They also show why an asymptotic correction must not be advertised as a finite certified bound: at this order its accuracy need not be symmetric in λ , and its remainder constant has not been optimized.

For the mathematical finite certificate, the main program evaluates (6.3) with $\vartheta = 1$, $J = \lfloor \sqrt{n}/\log(n+2) \rfloor$, and $M = \lceil 3N_n \rceil$. For $n = 64, M = 19$, it gives the numerical lower value 0.89482402, while the numerically computed ratio is 0.99453947. For $n = 256, M = 38$, the corresponding values are 0.90564333 and 0.99161984. Both calculations respect the proved inequality; neither uses outward rounding.

Prefix	n	Numerical ratio	Leading profile	First correction
$\delta_1 = 0$	10^4	0.62498322	0.59889287	0.62013748
$\delta_1 = 0$	10^8	0.62820473	0.60645535	0.62788283
$\delta_1 = 0$	10^{16}	0.62290548	0.60653066	0.62268465
$\delta_1 = 1/2$	10^4	0.63186838	0.60403388	0.62574393
$\delta_1 = 1/2$	10^8	0.62794894	0.60644250	0.62795760
$\delta_1 = 1/2$	10^{16}	0.62274259	0.60653065	0.62274334

Table 3: Retention ratios at critical coupling. Each model uses its own A, H_n, N_n . The first correction is (5.8).

M	$ML_M v_M$	$2M^2 \log(\rho_M/\rho)/A$	Reduced prediction	$\kappa_M/(AL_M)$
100	0.97577094	1.17856522	1.19057584	1.03909771
1000	0.99048085	1.13166298	1.13280358	1.01834924
10000	0.99469179	1.10170261	1.10181329	1.01065843

Table 4: Pure-tail finite-fold diagnostics. The reduced prediction uses the convergent solution $S(1/L_M)$ in (7.6). Its error is consistent with the separate algebraic remainder in $1/M$.

8.4 Reproduction and numerical limitations

The recorded environment used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The primary run uses fifty decimal working digits, and the supplementary run uses forty. The models implemented numerically are $a_j = j^{-3} + \delta_1 \mathbf{1}_{\{j=1\}}$, with $\delta_1 = 0, 1/2$; the proofs cover the larger class in (2.1). Fourier diagnostics truncate the scaled integration interval and the analytic series in the integrand. Their tails are not enclosed by interval arithmetic in the code. Exact finite comparisons and independent recurrence checks help detect mistakes, but working precision is not a certified number of correct digits.

From the package root, run

```
python -m pip install -r requirements.txt
python code/verify.py
python code/supplementary.py
sh build.sh
```

For a smaller numerical run, use `python code/verify.py -quick`. The article is self-contained and its typesetting does not require rerunning any computation or accessing the repository. All displayed numerical tables are recorded outputs rounded for readability.

9 Interpretation, proof boundaries, and formalization targets

9.1 Three uses of asymptotics, with different logical content

There are three distinct objects. Near $q = 0$, the solution is an ordinary convergent series, or equivalently an exponential transseries after $q = e^{-X}$. Near the critical boundary $q = \rho$, the inverse is a convergent analytic function of a finite logarithmic generating grid. Finally, as the sector index $n \rightarrow \infty$, the coefficients have a Gaussian–logarithmic asymptotic expansion. Convergence of the second object does not prove convergence or Borel summability of the third.

Similarly, $M \rightarrow \infty$ and $n \rightarrow \infty$ are not interchangeable without a scale hypothesis. The

finite-action folds exist for each M , but the exactly marginal boundary has no finite Puiseux chart. Their reduced logarithmic chart $S(1/L_M)$ is convergent; the theorem also keeps a separate, explicitly stated algebraic $1/M$ remainder. It does not claim that every term of a joint infinite $1/M$ -expansion converges.

9.2 Why the endpoint cutoff is especially efficient

The exact ratio can be written

$$R_n(M; 0) = \mathbb{P}(\text{no action exceeds } M \mid S_n = n).$$

At $M \asymp N_n$, forbidding large actions has a nontrivial Poisson cost $e^{-1/(2\ell^2)}$. Its effect on the mean of the normalized sum is only $O(H_n^{-1/2})$, because the fluctuation scale is larger by $\sqrt{H}$. Its effect on the normalized variance is $O(\log H/H)$. The normal local limit therefore does not add a new leading cost at the critical center. This yields a pure extreme-value profile, unlike a regime where individual large actions and the sum fluctuate on the same scale.

This explanation is made quantitative by the mean and variance shifts in Lemma 5.1. The cancellation of the prefix in the ratio requires comparing those shifts with the first correction to the full probability. It would be missed by retaining only the leading Gaussian. Thus the universality calculation is more than the leading extreme-value observation.

9.3 Dependency audit

The main proofs have a short, inspectable dependency structure. The exact coefficient formula uses formal Lagrange inversion and the analytic implicit function theorem near zero. The critical chart uses the local polylogarithm identity, another implicit-function argument, and an explicit contraction bound. The sector and cutoff theorems use Fourier inversion, elementary damping from positivity, Gaussian integrals, and Taylor estimates. The finite certificate uses only Fourier absolute values, a cosine inequality, and the Poisson shift identity. The fold drift uses analytic Riemann sums and a one-variable implicit equation.

No result depends on an unproved statement in the repository, an assumed resurgence theorem, a numerical fit, or a global complex singularity classification. Conversely, the article does not establish such a classification, an arbitrary-support Hahn-field closure theorem, a signed-weight analogue, or general transseries realization. This separation is necessary: a local analytic chart and a formal support statement are not automatically an asymptotic realization theorem for an unrestricted formal field.

9.4 A realistic Lean development

No new Lean file is supplied or claimed to compile. The inspected repository already has supporting asymptotic-scale, flatness, and well-based-support work, but that does not formalize the probability and analytic estimates of this article [1]. A useful implementation order is the following.

First formalize the finite coefficient recurrence, its equality with the exponential coefficient, monotonicity in the cutoff, and exact equality through degree M . These are finite algebraic statements and are the closest match to the exact checks supplied here. Next isolate the Poisson shift identity, finite-cutoff conditioning identity, and elementary anti-concentration majorant. Together they prove the finite certificate without requiring the full asymptotic theory.

The analytic stage should expose the hypotheses of the local chart as explicit norm bounds and use a quantitative contraction interface. A separate Fourier stage should package uniform damping, integrable Taylor remainders, and the large-scale limits. Only after these pieces are formalized should the prefix-cancellation and minimal-budget theorems be assembled. This division prevents a verified finite coefficient identity from being misreported as a verified infinite asymptotic theorem.

10 Further research questions and concrete targets

1. A uniform transition as the stable exponent approaches two. For $a_j = aj^{-1-\alpha}$, let $\alpha \uparrow 2$ while $(2 - \alpha) \log n$ remains bounded. Construct a joint inverse chart and coefficient expansion that displays the cancellation between the singular gamma factor and the quadratic analytic term. Determine how the stable cutoff density crosses over to $e^{-1/(2\ell^2)}$, and whether the prefix-independent first correction has a uniform interpolating form. Separate limits in α and n do not answer this question.

Editorial note (ProveIt, 2026-09-29). Answered, pending review, by two later packages of this series written independently of each other, both from a library copy of this article. The stable-Gaussian package [7] targets this question in exactly this window, from both sides, and gives a joint inverse chart about a desingularized core, a two-term uniform inverse, the coefficient law with its first correction, the cutoff law $\exp(-\ell^{-\alpha}/\alpha)$ that crosses over to $e^{-1/(2\ell^2)}$, and a uniform prefix-independent first correction, which at $\alpha = 2$ is (5.8). The confluent package [6] allows $2 - \alpha_n \rightarrow 0$ at any rate and of either sign, with one convergent chart in three variables and a leading-order coefficient and cutoff law. Where both apply they agree term by term.

2. Slowly varying perturbations of the marginal tail. Replace the exact tail by $a_j \sim aj^{-3}L(j)$, with a specified slowly varying function L . The sum scale is governed by a truncated second moment, while the cutoff scale is governed by $n \sum_{j>M} a_j$. Find conditions under which their ratio diverges, remains bounded, or has an explicit logarithmic law. Identify classes of L that preserve a finite convergent generating grid, and classes that require genuinely new inverse notation or de Bruijn conjugates.

Editorial note (ProveIt, 2026-09-29). Partial progress: the logarithmic-endpoint package [5] treats $L(j) = (\log j)^r$, $r \geq -1$, at leading order (its `thm:weightedchart`, `thm:weightedbudget`). General slowly varying L remain open.

3. Higher cutoff corrections and the return of prefix data. Compute the full expansion in $H^{-1/2}$ and $\log H/H$, with a systematic Gaussian-logarithmic coefficient calculus. Determine the first order at which finite-prefix information survives in the normalized ratio, if it does. The present cancellation establishes only the displayed first correction; it is not evidence that every higher order is universal. A useful deliverable would be a symbolic algorithm with an explicit uniform remainder for each fixed requested order.

4. Large-order growth of the logarithmic sector expansion. Analyze $G_k^{(b)}(\lambda)$ as $k \rightarrow \infty$, uniformly on useful λ -sets. Determine least-term truncation, exponentially improved remainders, and possible Borel or accelerated summation. The finite formula (4.13) provides a concrete starting point, but its existence alone does not establish a Gevrey class or resurgence. The relation between the convergent critical chart and this potentially divergent outer expansion deserves a separate theorem.

5. Growing coupling windows and moderate deviations. Find the largest range $|\lambda| \leq L_n \rightarrow \infty$ on which a relative, rather than additive, version of Theorem 4.1 holds. Match it to subcritical one-large-action behavior and supercritical saddle behavior. The division by $\varphi(\lambda)$ in the cutoff correction makes clear why compact-window estimates cannot simply be extended without new analysis.

6. Joint precision and action-budget limits. Let $\eta = \eta_n \downarrow 0$ with n . Determine the uniform range of (5.12), and the sharp relation between the asymptotically minimal budget and the certificate budget. The successive-limit statement in Corollary 6.3 does not settle this joint limit. Develop an adaptive choice of J, ϑ that improves the finite majorant without losing a simple, checkable expression.

7. Interval-certified coefficient extraction and bit complexity. Implement the positive recurrence with directed rounding and explicit special-function enclosures, producing a machine-readable retention certificate. Then investigate faster coefficient algorithms, block convolution, and reliable rescaling for very small normalized coefficients. Compare their bit complexity with the arithmetic bound $O(nM)$. A useful formal target is a small checker for the final inequality, independent of the heuristic algorithm used to choose M .

8. Vector feedback and noncommensurate actions. Replace the scalar feedback by a positive multitype system or a finite number of coupled marginal tails. Determine whether the covariance matrix has a logarithmically divergent subspace and how the extreme-action budget interacts with the remaining finite-variance directions. Noncommensurate actions require a replacement for integer coefficient extraction; the appropriate object may be a Laplace measure rather than a single power series. This is not a routine change of notation.

9. Signed weights, complex saddles, and analytic continuation. Allow finite or controlled infinite sign changes while preserving a chosen analytic small solution. Determine which cutoff conclusions survive without coefficient monotonicity or a positive Poisson representation. Even for the positive model, describe global continuation of the Lambert-core chart and the singular geometry encountered away from the real marginal boundary. The local pole $w = 1$ in coefficient functions need not by itself identify an actual singularity of the full solution in a different parameter regime.

10. Formalization of the finite-to-asymptotic bridge. Prove the finite certificate first in Lean, then connect it to a separately formalized marginal local limit theorem. An important interface question is how to represent compact-uniform Fourier bounds and analytic logarithm branches so that the analytic hypotheses remain visible in downstream statements. The final artifact should distinguish a certified numerical retention ratio from a formally proved asymptotic budget law.

11 Conclusion

The marginal j^{-3} feedback class lies between ordinary finite-action folds and nonintegral stable criticality. Its logarithm is not a defect to be suppressed: an exact Lambert- W_{-1} core turns it into a convergent, explicitly controlled inverse chart. At the coefficient level, a different logarithmic expansion describes a Gaussian sum with infinite second moment.

The principal cutoff result separates the extreme-action scale $\sqrt{n}$ from the larger fluctuation scale $\sqrt{n \log n}$. It identifies the exact limiting retained fraction, its first prefix-independent correction, and the first correction to the minimal budget. A finite positive inequality then supplies a logically independent route to certified truncation. Finally, the moving finite folds reproduce the same leading action cost through their critical-value drift, with their own convergent reduced chart.

These results constitute a proved model-specific extension of the inspected repository program. They are not a claim of established publication priority or a resolution of unrestricted transseries realization. The explicit coefficient, support, and error formulas make the next steps—higher corrections, uniform endpoint crossover, and proof-producing computation—precise enough to test and formalize.

References

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